

# Qatar-1b

R-2701

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_110902 · created 2026-09-01T00:19:45+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2455806.67644 +/- 0.0004 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.189 +/- 0.056                  |
| Transit depth (Rp/Rs)^2                           | 3.58 +/- 2.13 [%]                |
| Orbital Inclination (inc)                         | 82.16 +/- 2.96                   |
| Airmass coefficient 1 (a1)                        | 0.119 +/- 0.063                  |
| Airmass coefficient 2 (a2)                        | 1.67 +/- 1.47                    |
| Scatter in the residuals of the lightcurve fit is | 51.1 %                           |
| Best Comparison Star                              | #1 - [91, 28]                    |
| Optimal Aperture                                  | 3.02                             |
| Optimal Annulus                                   | 7.66                             |
| Transit Duration (day)                            | 0.053 +/- 0.028                  |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.189 ± 0.056 vs 0.1463 (deviation 0.67σ) |
| Duration fit vs published | 0.053 d vs 0.0692 d (deviation 0.51σ)     |
| Mid-time O–C              | 1.0 min                                   |
| Depth SNR                 | 3.0                                       |
| Residual scatter          | 51.1 %                                    |

### Light curve

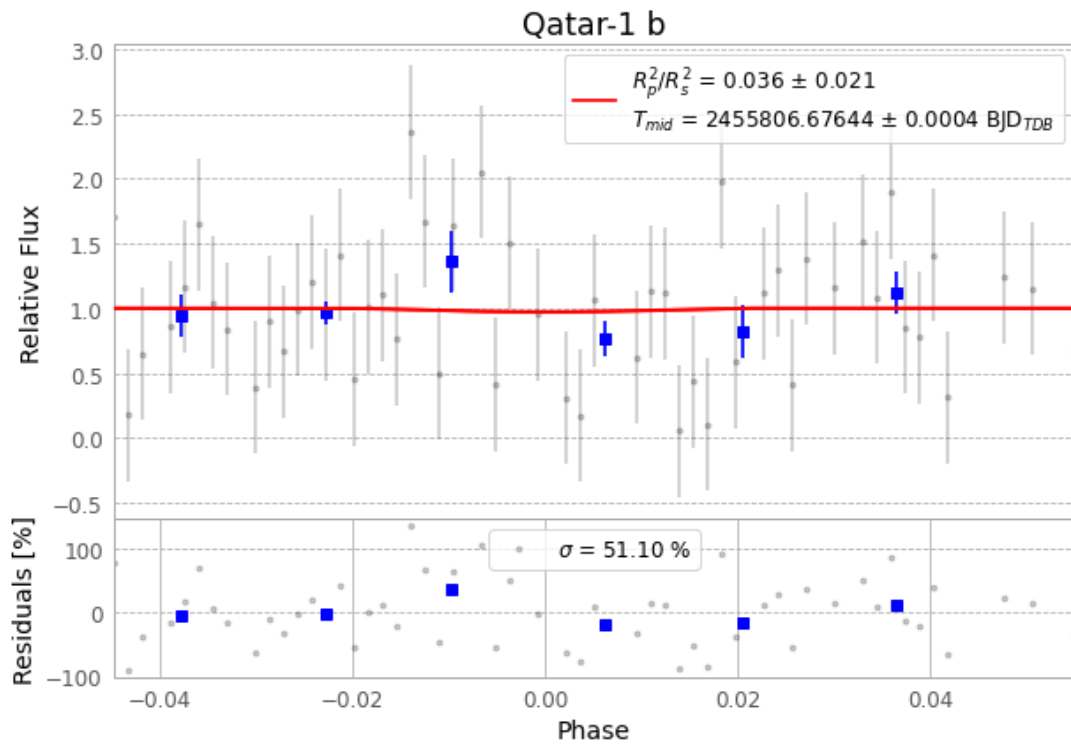


Figure 1 — FinalLightCurve\_Qatar-1 b\_2011-09-01.png (SHA-256 09f30501506d3ff1...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [277, 422], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 9f8f081624aece31...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

73 FITS frames (48 MB), Qatar-1110902022712.FITS → Qatar-1110902060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110902.log (1e864d42851a...), FinalLightCurve\_Qatar-1 b\_2011-09-01.png (09f30501506d...), FinalLightCurve\_Qatar-1 b\_2011-09-01.csv (38d0b1fbb54a...)

## Compute resources

Wall time 164.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:19Z.